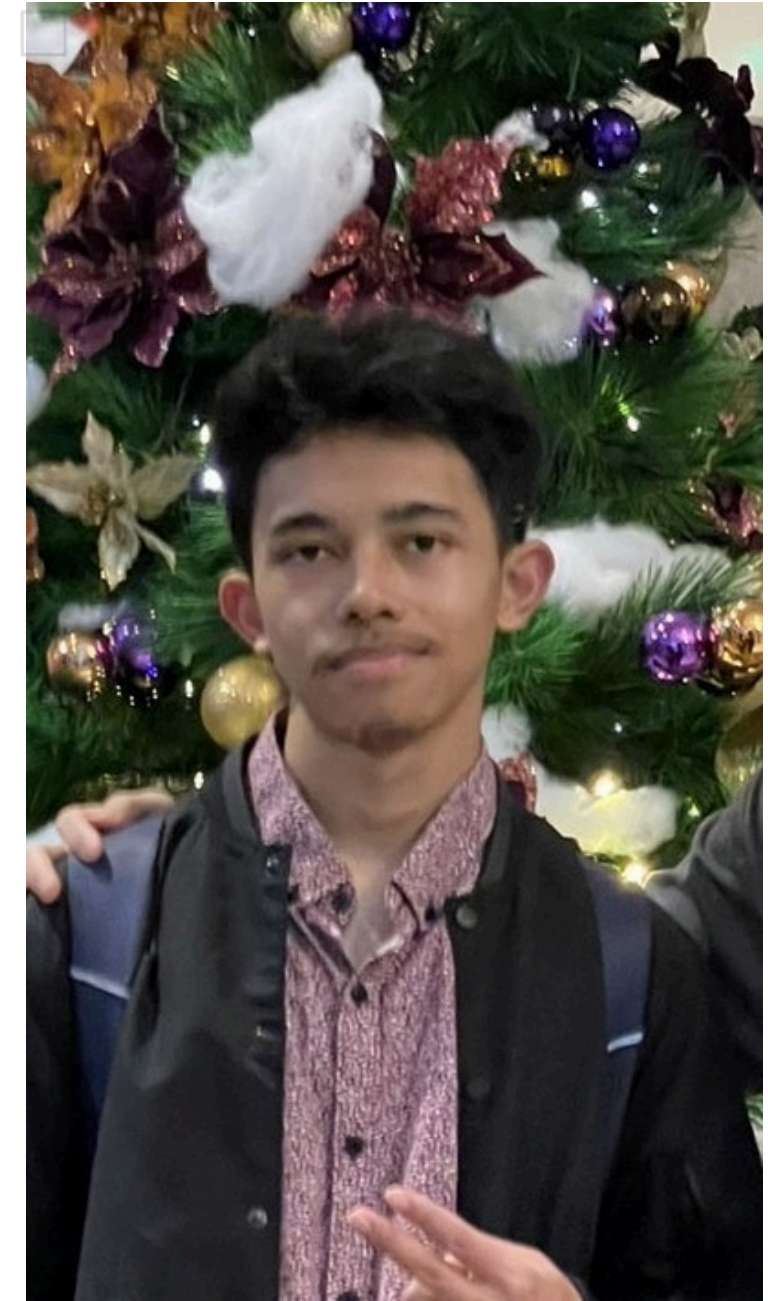


Hi, Im Frans

3rd year EECS Student at NTHU.

Aside from being a student, I am an electronic hobbyist, programming enthusiast and enjoy working with others.



Experience

School Club - NTHU Racing

Electrical system - member



Figure 1 : Presentation



Figure 2 : Test-drive

Competition : MoE 2024 Mobile Communication Practical Competition

3 month project :

Built a smart agricultural drying system

Responsibility : Firmware & sensors integration



Figure 1 : Debugging



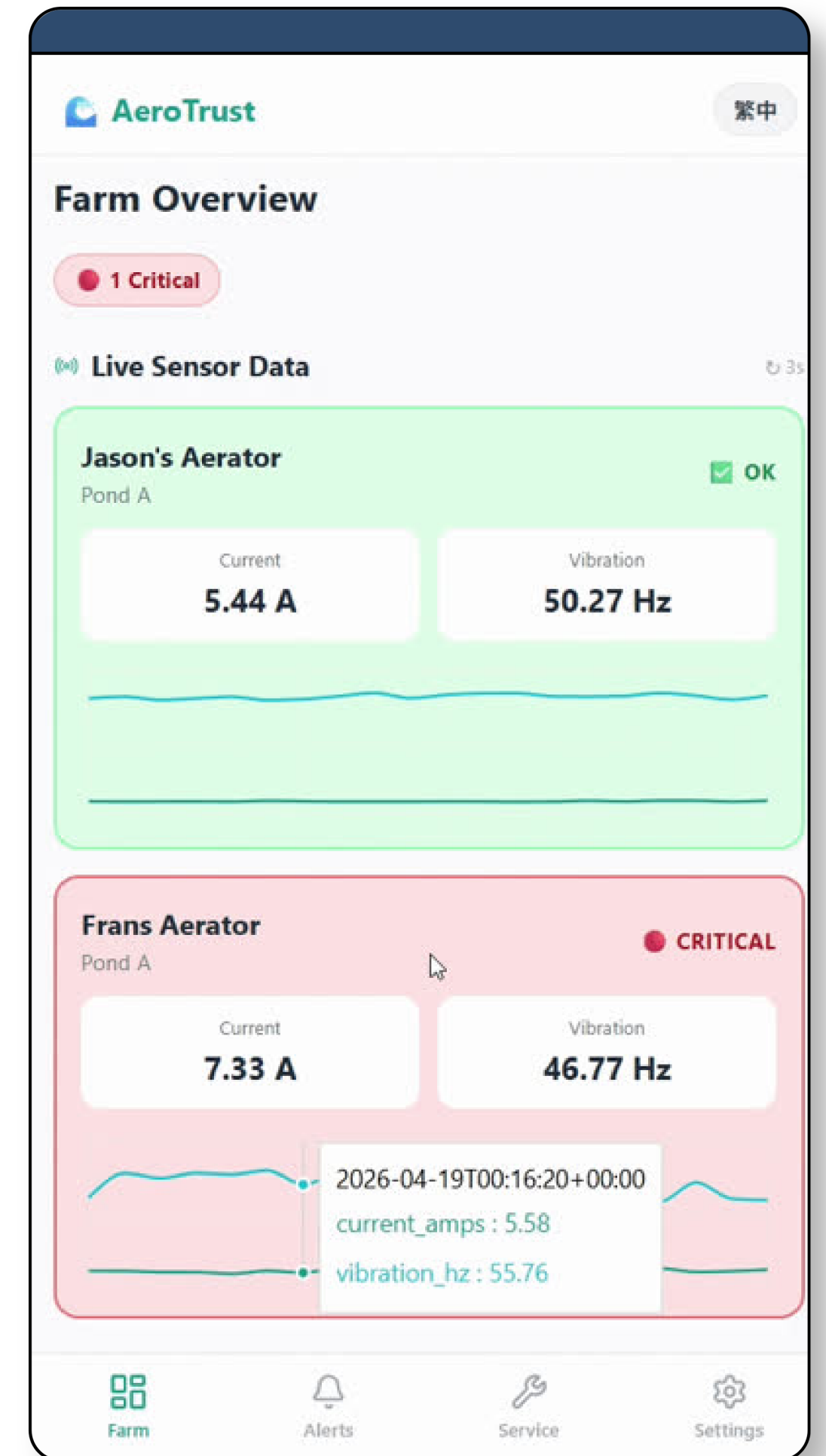
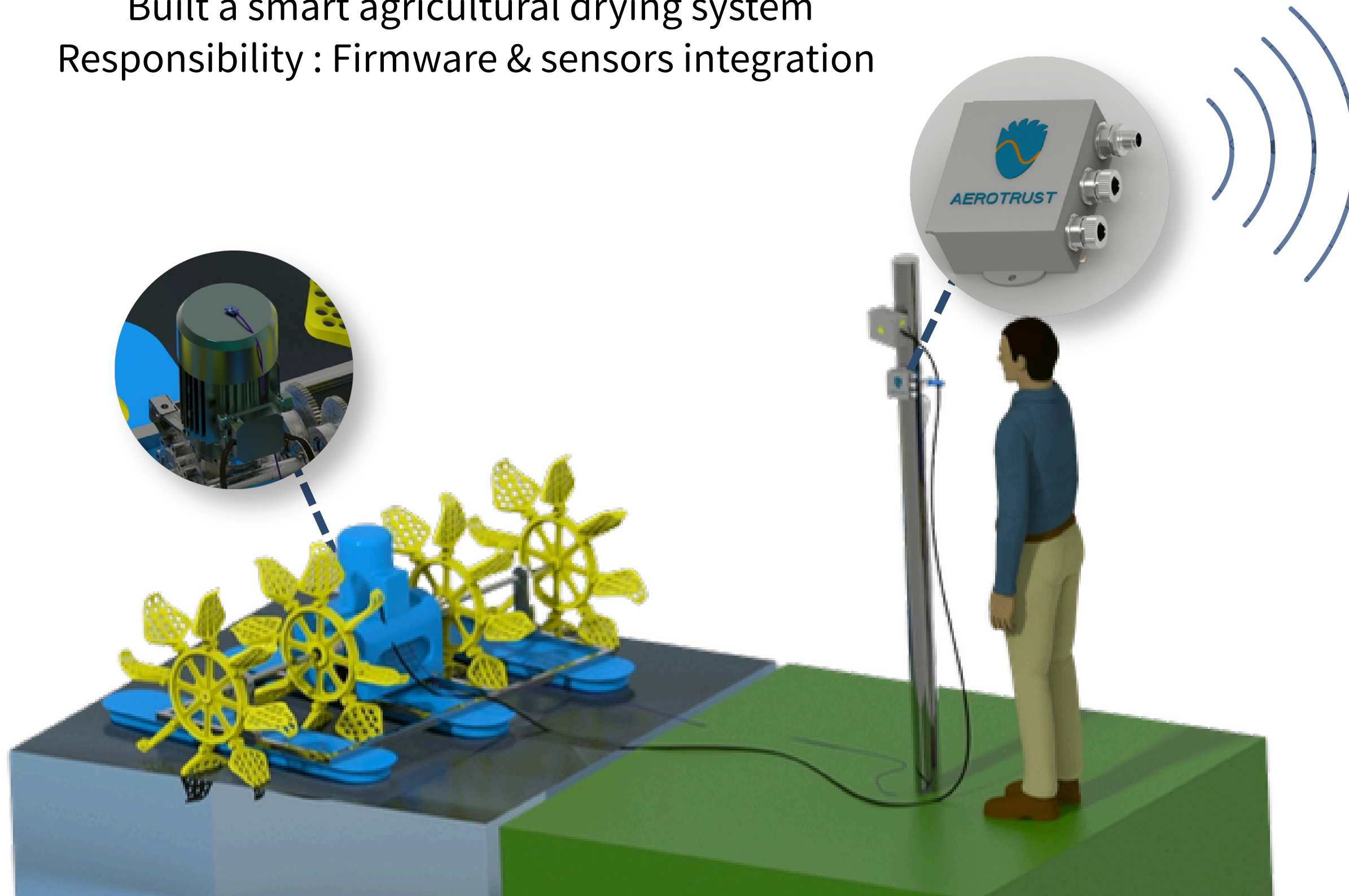
Figure 2 : 1st Place winner award

Aerator monitoring System

5 month project :

Built a smart agricultural drying system

Responsibility : Firmware & sensors integration



HULT Prize Taiwan 2026 Nationals Winner

Represent Taiwan in the Digital Incubator Hult Prize globals 2026



Figure 1 : HULT Prize (QNA Session)



Figure 2 : HULT Taiwan Nationals Winner (Team Cultivator)



Figure 3 : Accepted to NTHU Garage accelerator (AeroTrust)

International Student Association

Event Planner



Figure 1 : Host speaker



Figure 2 : Group photo

Side Projects

NTHU RACING project : Interactive pop-up generator for pdf/image file

see how it works



Product Integrations Sign In

Upload a PDF or Image file

Turn your image into an interactive, flexible .html file
store your .html as link, host it for further use!

Start now

<https://popupio.onrender.com/>

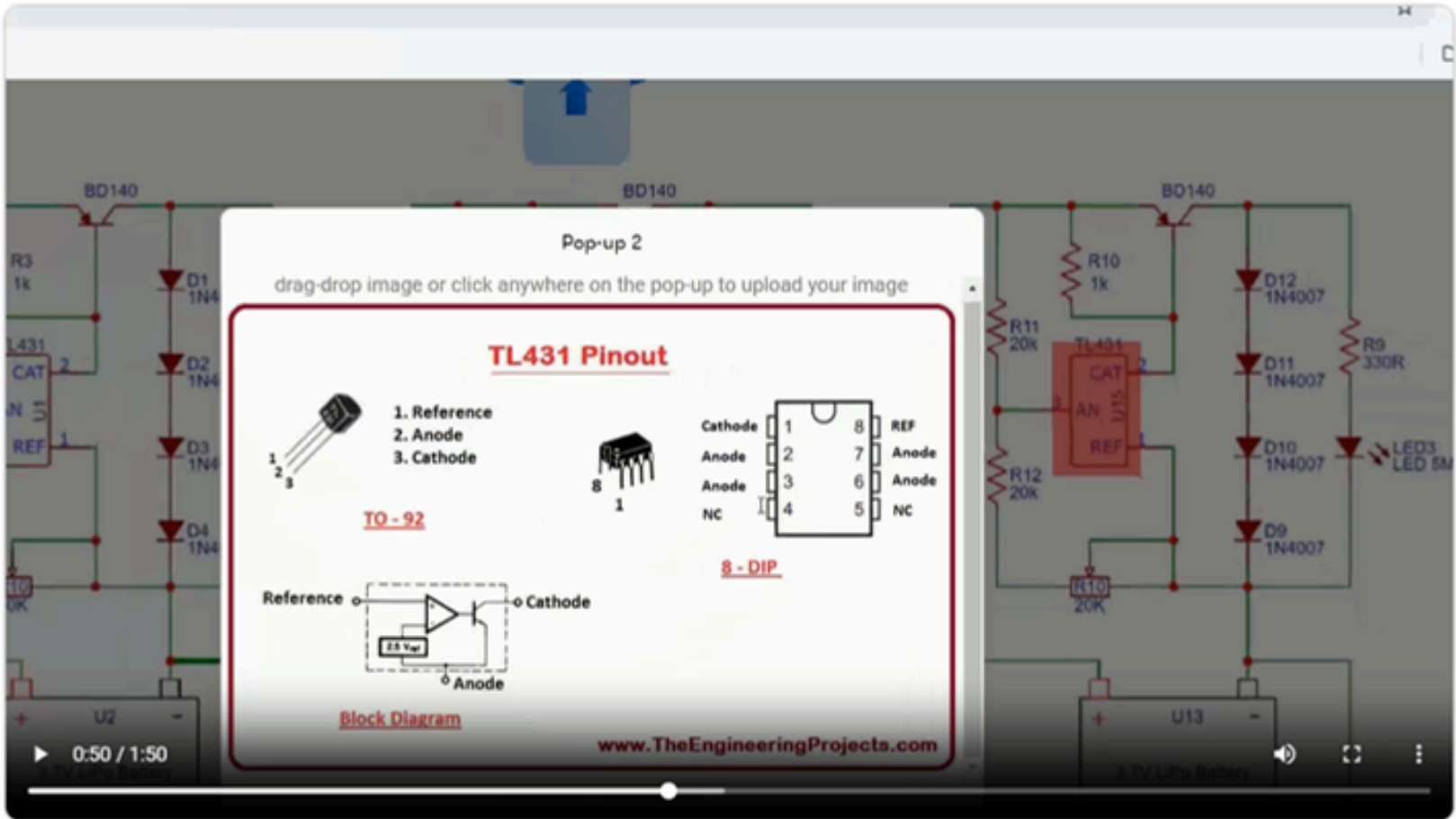


Figure 2 : Demo

FPGA Audio Visualizer

2 FPGAs (Basys-3)
Reactive light component +
memory (store light sequence)

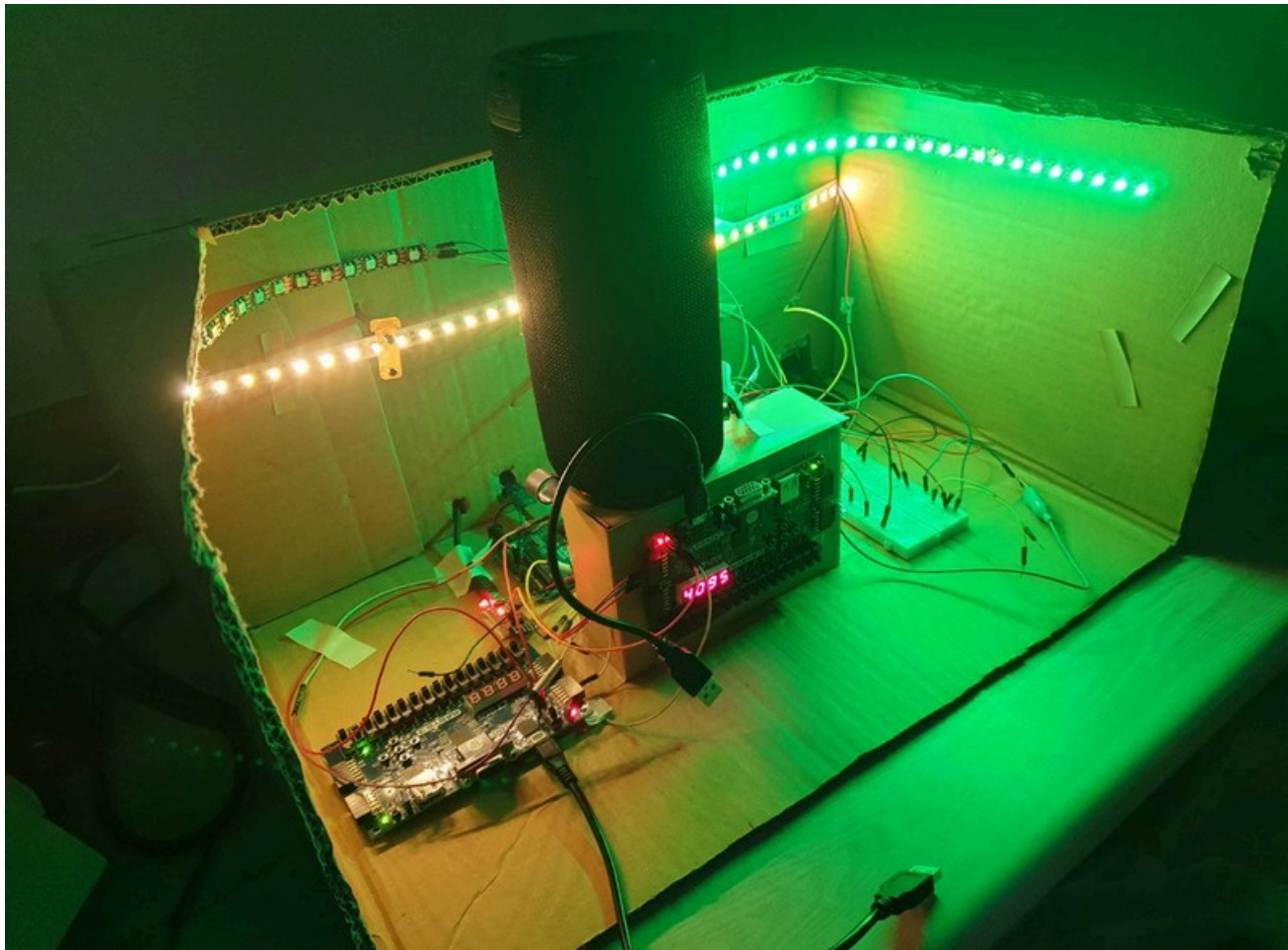


Figure 1 : Final demo

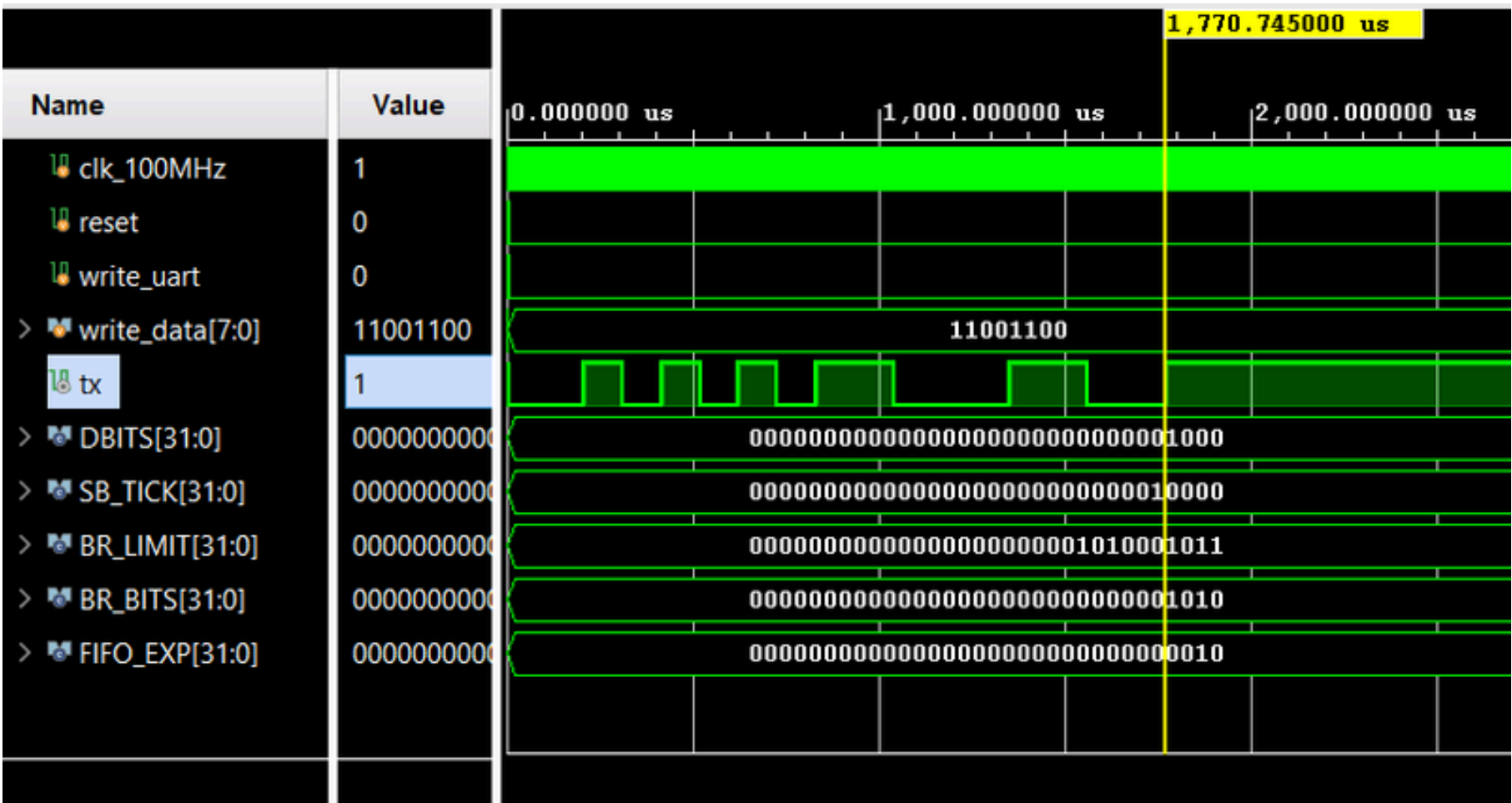
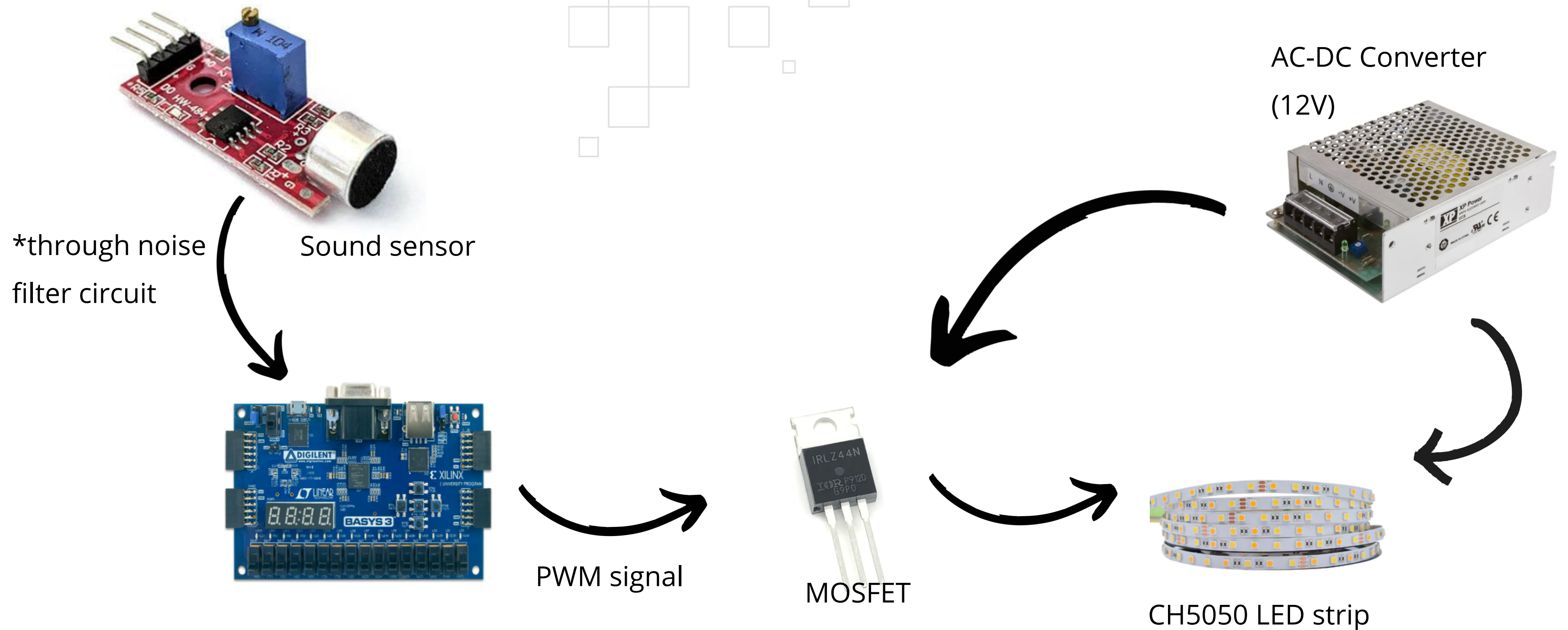


Figure 2 : UART TX Simulation

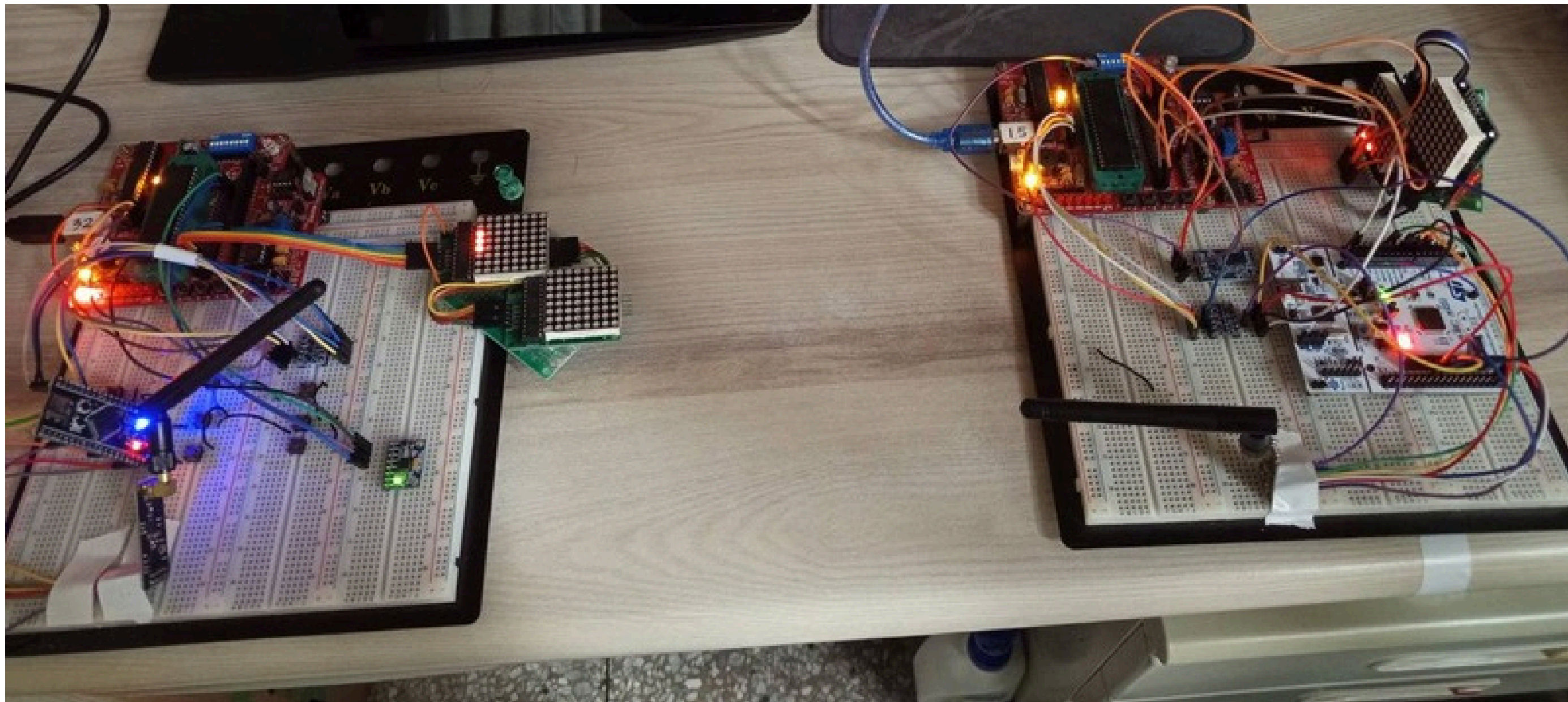
FPGA – RESPONSIVE LIGHTING

- Uses sound sensor and Vivado XADC to convert analog signal to digital signal.

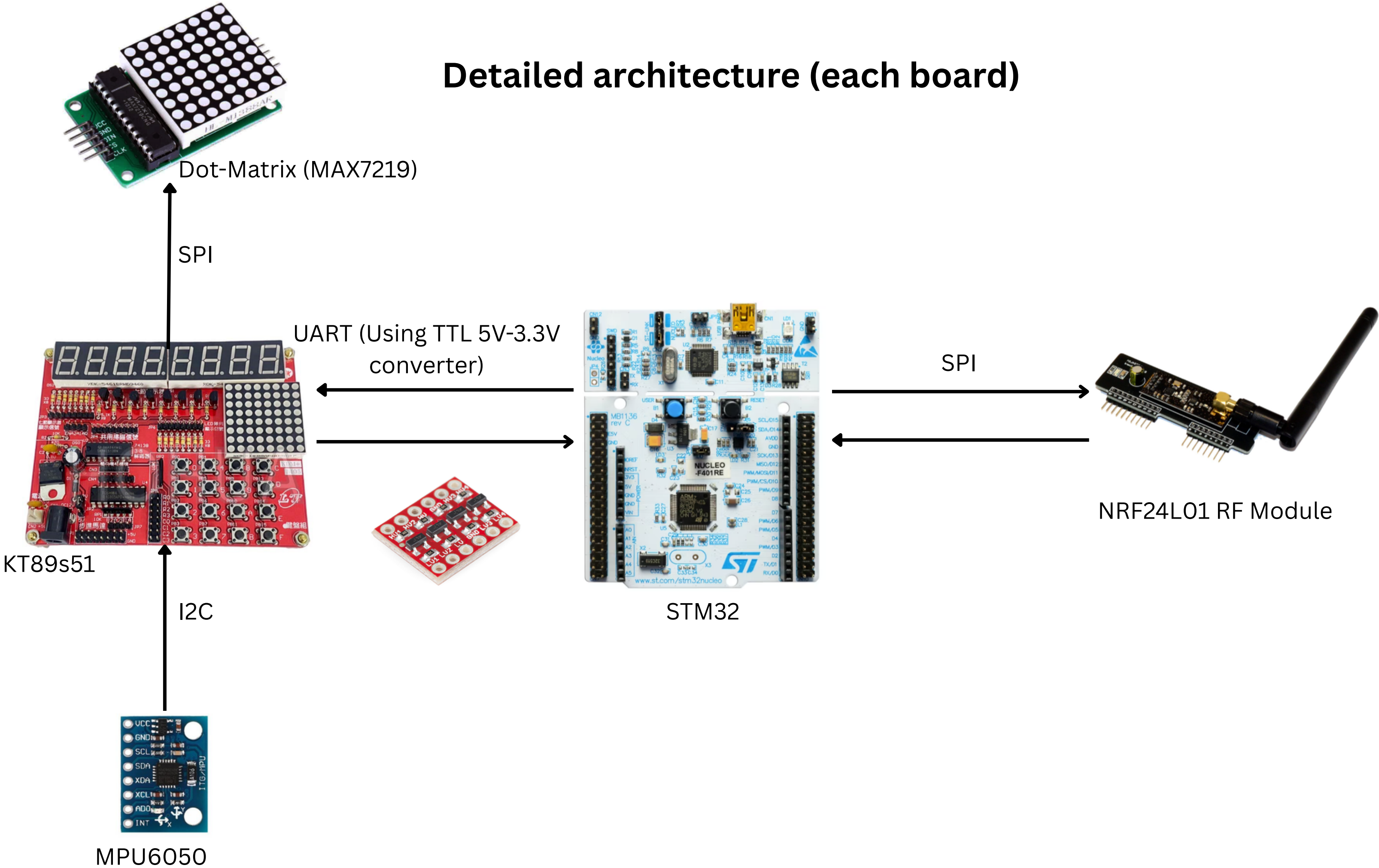


Wireless multiplayer Pong

Bare-metal programming and
driver development



Detailed architecture (each board)



Matlab + Plecs (Power electronic special topic)

Various DC-DC converter topologies simulation

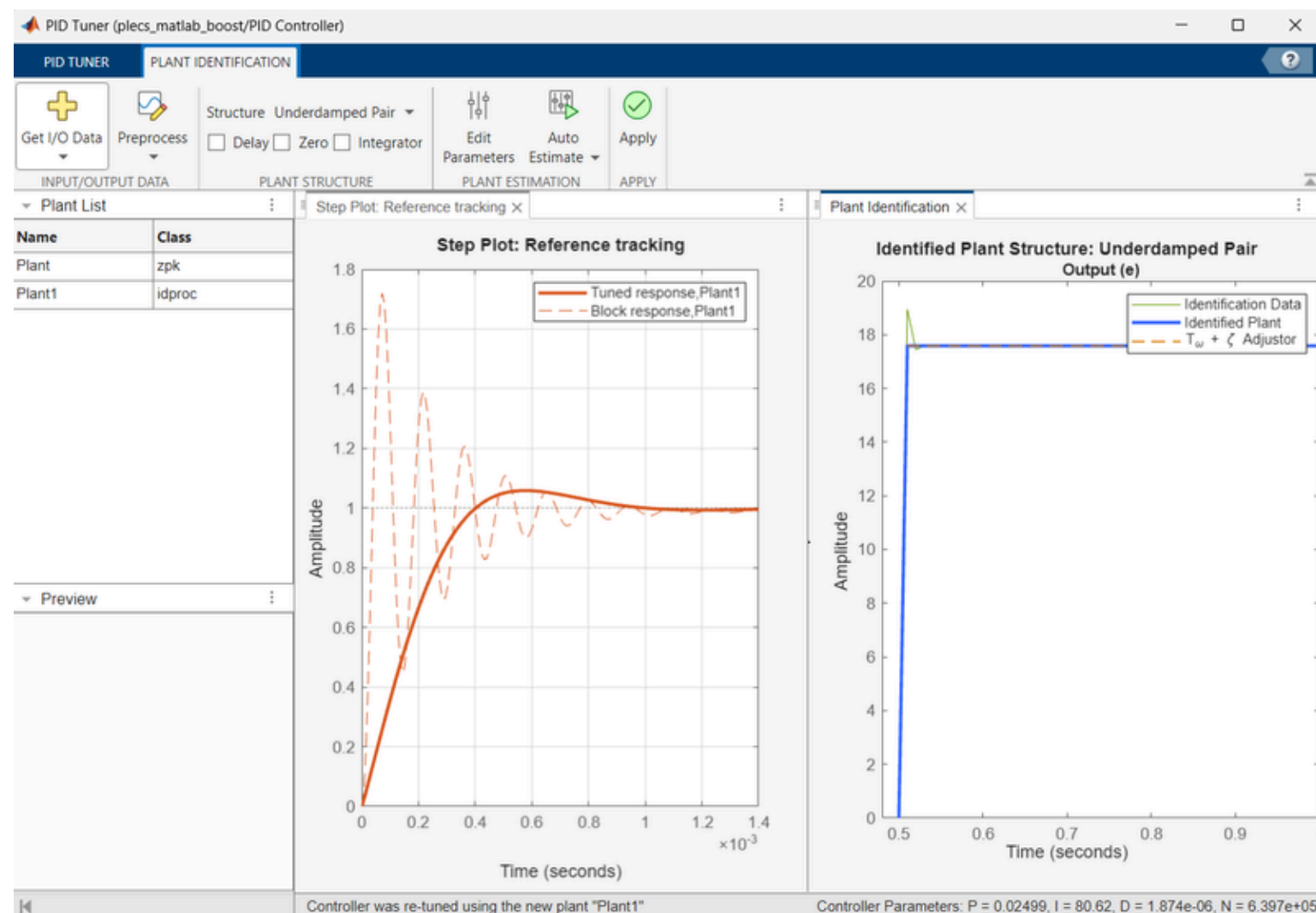


Figure 1: PID Tuning

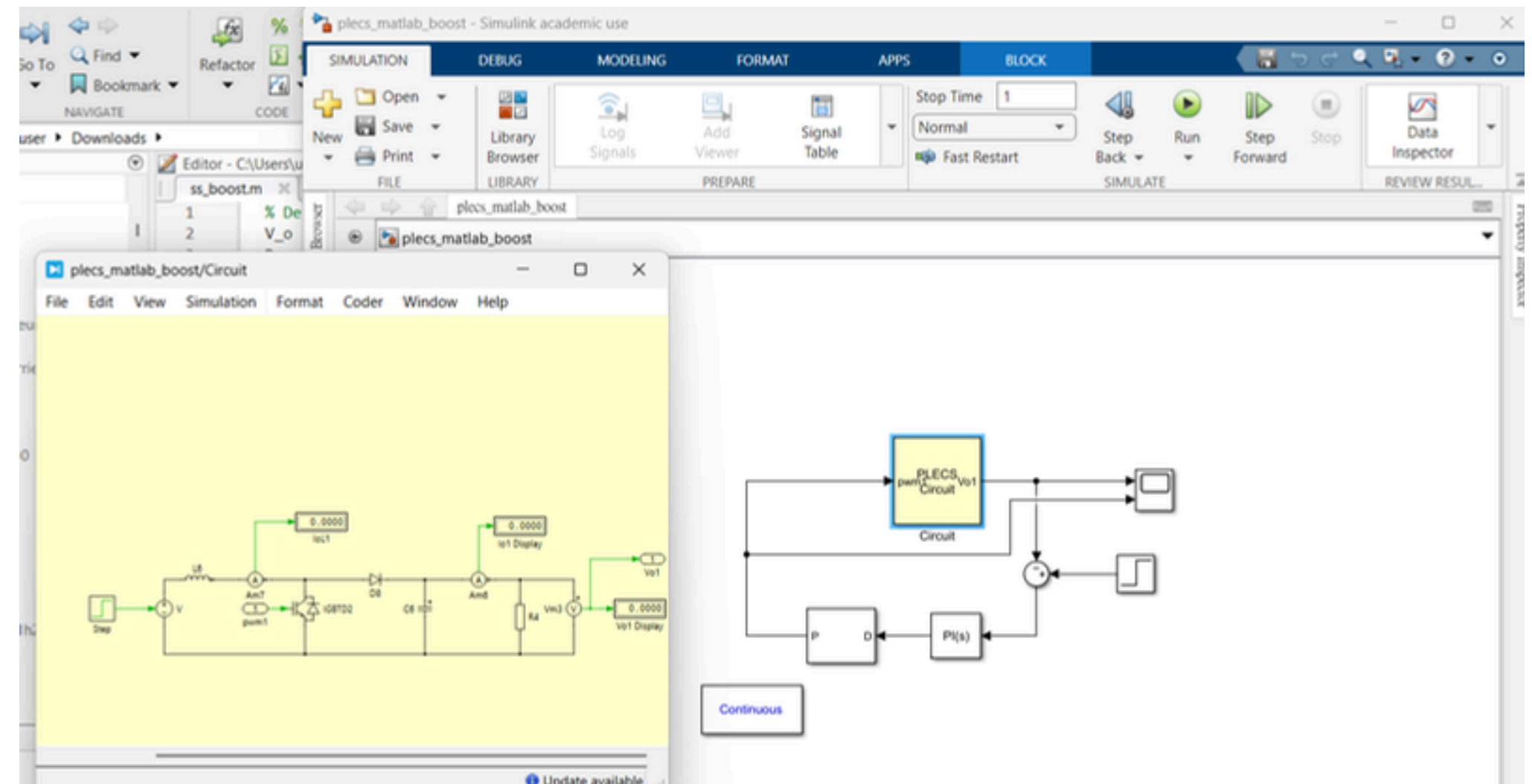


Figure 2 : Plecs Block-set

ROM access controller (RAC)

Include : Frequency divider, 8x8 array Read-out circuit and an Overlapping Detector

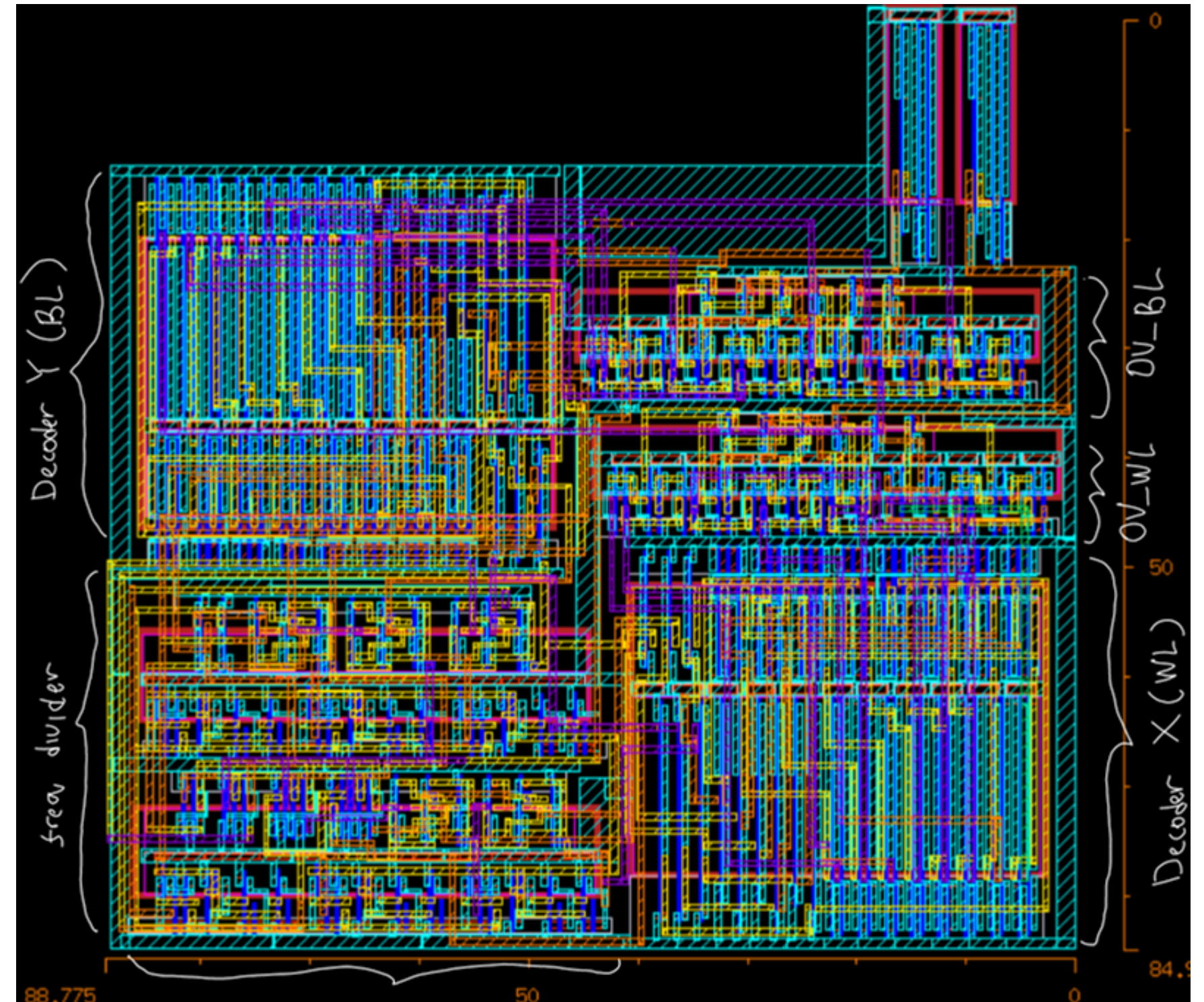
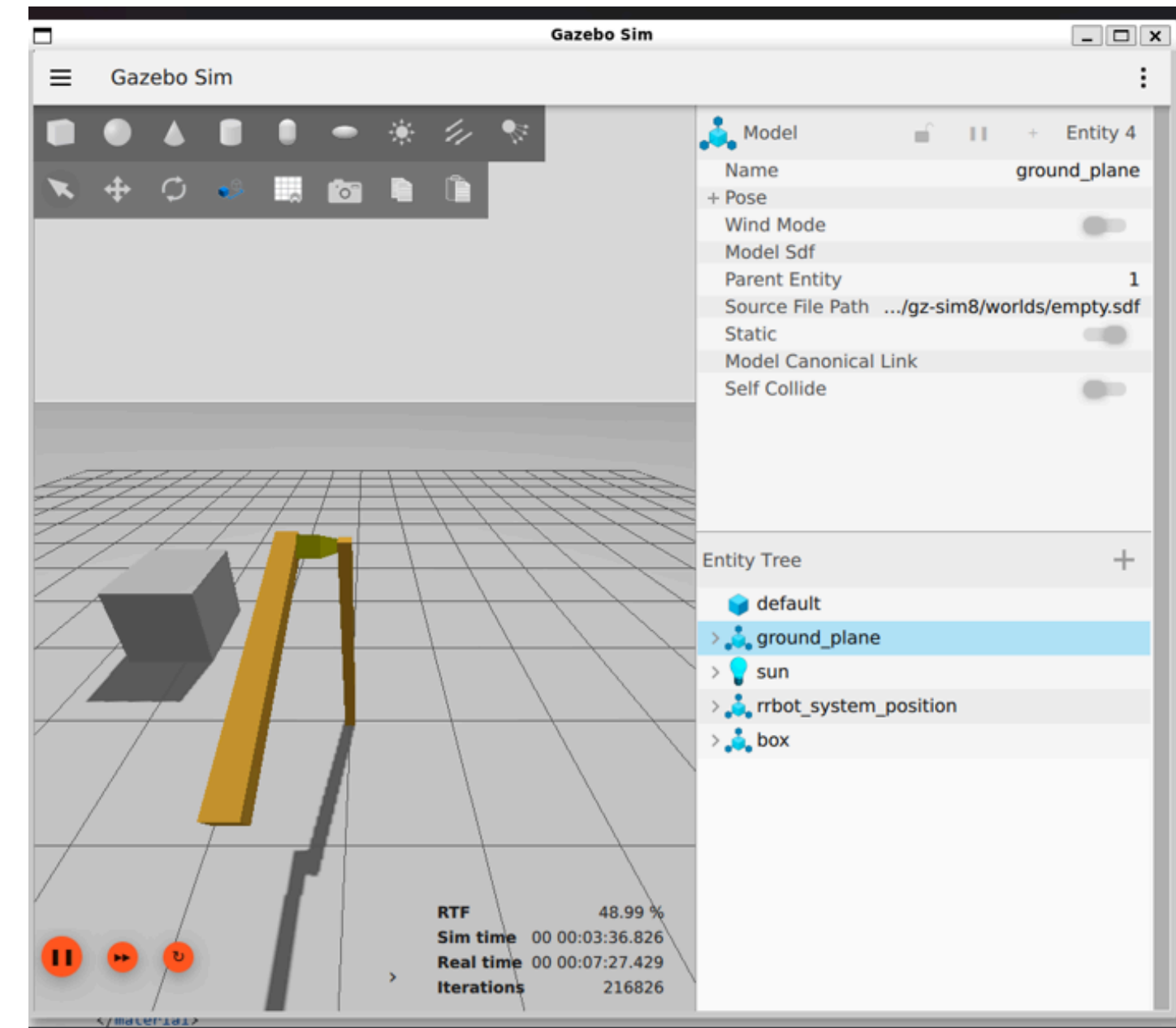
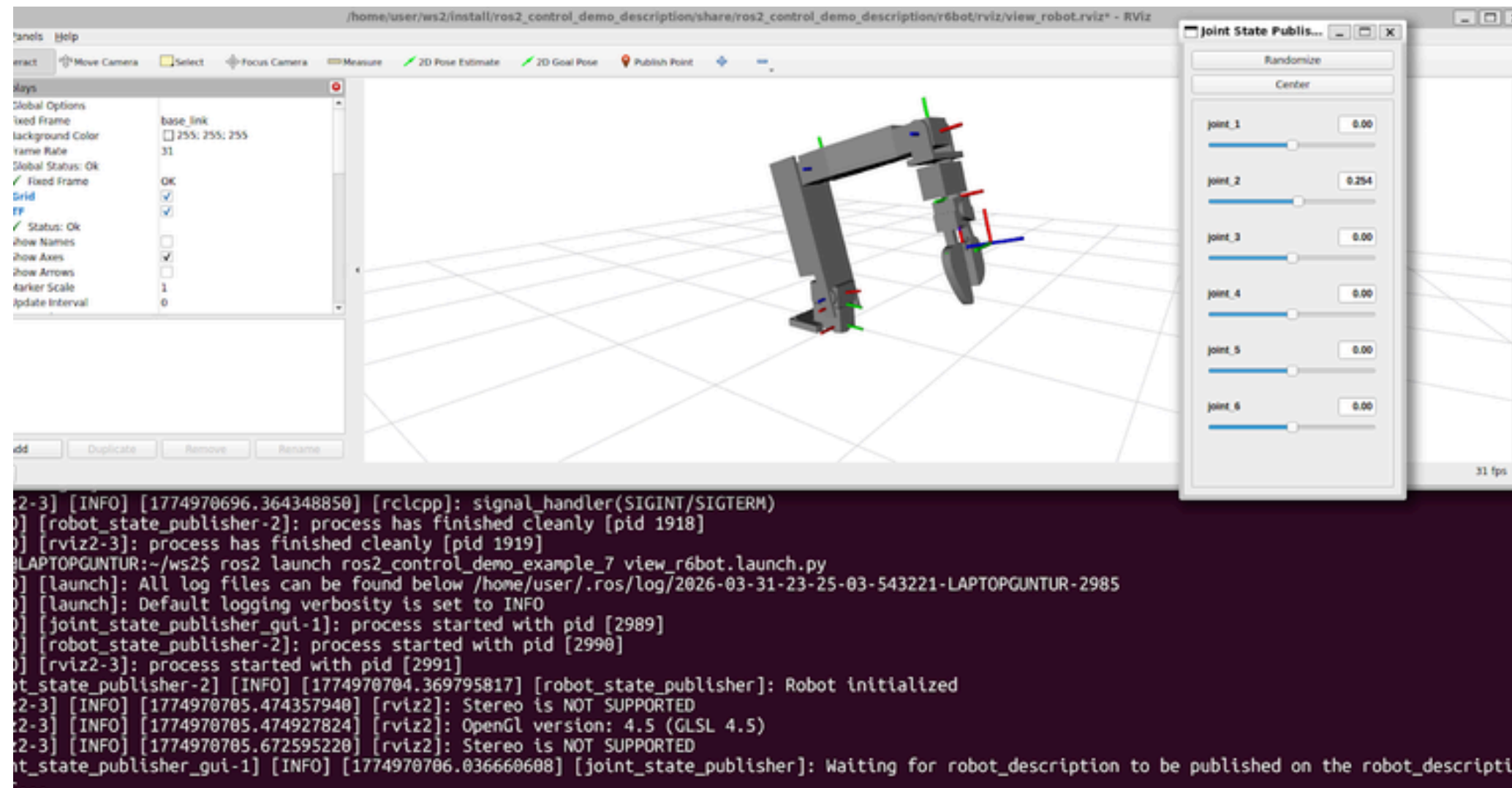


Figure 1 : Final layout

6DOF Robotic arm Simulation

- ROS2 - Rviz
- Using common ros2 nodes/packages



Thank you!